

Munaza Ashraf

Applied ML Engineer and Researcher
World Models and Physical AI

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Education

- Sep 2020 **B.Sc. (Hons.) Software Engineering**, *University of Engineering and Technology (UET)*
Jun 2024 *Taxila, Pakistan*
- CGPA **3.94/4.0**, **Silver Medalist** (second highest aggregate across all four years).
 - Final Year Project: PsycheSync, an AI mental health platform built on fine tuned LLMs.

Research Interests

- Areas** World Models, Embodied and Physical AI, World to Policy Distillation, Efficient Multimodal Learning, Robot Learning
- Direction** Building **world models for physical AI** that capture how the world moves and changes, and are small and fast enough to run on real robots. My focus is knowledge distillation, which moves the spatial, temporal, and physical understanding of large multimodal models into compact models and policies that fit tight latency, memory, and compute budgets.

Publications & Preprints

- 2026 **Recovering a Withheld Audio Tokenizer from Its Generator: Reverse Distillation of the MiniMax Music 3 Encoder, In and Out of Distribution**
Under Review *Munaza Ashraf, Bghira. Under review, 2026.* [Code] [Models] [Training Data] [MM3 OOD]
- Recovered the audio tokenizer that MiniMax Music 3 never released, using reverse distillation, so the 8B text only model can now take audio prompts for reference guided generation and continuation.
 - Trained only on generated music, the encoder **works just as well on real recordings (0.92 vs. 0.91)**, and training on degraded audio fixes lossy input (**0.82 vs. 0.50**).
 - Released the encoders, code, and MM3 OOD, a new **240** track test set.

Independent Research & Reproductions

- Aug 2026 **Reimplementation: WPT**
Sep 2026 *World-to-Policy Transfer via Online World Model Distillation.* [Paper]
- Rebuilt WPT on nuScenes with a frozen pretrained **OccWorld** teacher instead of the paper's Drive-OccWorld, covering the full loop from world model rollouts to the distilled policy.
- May 2026 **Reproduction: HASTE**
Head Wise Adaptive Sparse Attention for Video Diffusion. [Paper]
- Reproduced this training free sparse attention on Wan 2.2 and LTX 2.3, reusing masks across timesteps with an error guided top p budget per head.
- 2026 **Low Latency Content Safety Classifier, Runware**
[Base Model]
- Fine tuned Freepik's EVA based ViT to flag sexual content by severity, suspected sexualisation of minors, graphic violence, and extremist imagery.
 - It checks every generation request in production and returns a confidence score per category in **under a second**.
- 2025 **Independent Benchmark: Diffusion Inference Caching**
2026
- Compared FBCCache, TeaCache, MagCache, Cache DiT, and Fast3D Cache on image, video, and 3D models, weighing time and VRAM saved against FID, CLIP, SSIM, and LPIPS.

- May 2026 **Efficient Fine Tuning and Optimizer Studies**
- Jul 2026
- Tested TREAD token routing with LoRA and context parallelism, and compared SOAP, Muon, AdamW, and Lion on diffusion, LLM, and VLM adapters under matched settings.
 - Designed segmented gradient checkpointing for Flux and MageFlow, which gives finer control over memory versus recompute and lowers step latency.

Research and Work Experience

- Nov 2024 **Applied ML Engineer, Runware, London, UK (Remote)**
- Present
- Tools:** PyTorch, Diffusers, Transformers, vLLM, Go, CUDA, ROCm, NCCL, FlashAttention, SageAttention, FlashInfer, Docker, Prometheus, Vercel
- Integrated, optimized, and served **1,000+** production models across image, video, audio, 3D, LLMs, and VLMs, with up to **2×** faster inference and a scheduler per model family.
 - Built distributed and serverless inference on NVIDIA Hopper, Blackwell, Ada, and Ampere GPUs (H100, H200, B300, L40S, RTX PRO 6000, RTX 4090, RTX 3090) and AMD ROCm, using context parallelism and overlapping NCCL communication with compute.
 - Benchmarked FlashAttention, SageAttention, FlashInfer, and cuDNN SDPA on each GPU generation so every model runs on its fastest attention backend, alongside diffusion and KV caching and FP8, FP4, INT8, and INT4 quantization.
 - Designed and evaluated **teacher–student knowledge distillation** for smaller, lower latency model variants.
 - Trained LoRAs once on a base model and deployed them on its distilled and turbo variants by skipping or bridging the layers that shift most.
 - Built Runware's LoRA training platform and the Go microservices that serve vLLM models, with LLMops monitoring.
 - Supported high volume enterprise customers including Higgsfield, Freepik, Wix, PixVerse, Civitai, and the LTX team, with integrations deployed on Vercel.
- Sep 2024 **Founding Engineer, MedTalk, Faz Australia, Australia (Remote)**
- Jul 2025
- Tools:** Whisper, AssemblyAI, MedPaLM, LLaMA, RAG, Pinecone, WebSockets, AWS, RunPod
- Owned ML, backend, and infrastructure as a founding engineer of MedTalk, a real time clinical AI platform used by **50,000+** clinicians.
 - Built clinical training datasets and fine tuned speech, medical NLP, and language models, including MedPaLM, to write structured clinical notes.
 - Cut streaming speech latency after VAD from **~150 ms to ~50 ms P50 (3×** faster) through WebSocket streaming and better model serving.
 - Built a RAG based clinical assistant and a retrieval augmented billing code LLM on Pinecone and AWS.
- Jun 2024 **Computer Vision Engineer, VisionRD, Islamabad, Pakistan**
- Nov 2024
- Built a wearable AI assembly copilot for Toyota (head mounted 4K camera, real time error detection, cycle analysis), improving line efficiency by **~35%**.
 - Built worker activity tracking and assembly checks for Thal Engineering, and visual defect detection for KIA Motors.
- 2022 **STEM Trainer, Junior Coderz, Islamabad, Pakistan (Remote)**
- 2024
- Taught Scratch, Python, app development, and robotics to students in Canada, the USA, and the UAE, and prepared students for USACO, Bebras, and AI project work.
- Mar 2023 **Machine Learning Intern, NESCOM, Islamabad, Pakistan**
- Oct 2023
- Developed a deep learning ad detection and classification system, from data collection to evaluation.
- May 2022 **Deep Learning Intern, Bytewise Limited, Islamabad, Pakistan**
- Sep 2022
- Trained computer vision and NLP models in TensorFlow and PyTorch with custom datasets and augmentation pipelines.

Selected Projects

- Aug 2026 **Juricratic: AI Litigation Intelligence, Personal Venture, with Connor, juricratic.com**
- Present
- Treats a lawsuit as an imperfect information game and simulates it with LLM agents acting as judge, jury, opposing counsel, and client.
- Mar 2026 **Padelytix: Real Time Multi Camera Sports Perception**
- Aug 2026
- Tracks players and the ball across fixed cameras, reidentifies players by clothing, and cuts highlights, even with occlusion and out of sync cameras.

- Apr 2025 **Unilever × PSL 2025: Broadcast Logo Detection and Sponsorship Analytics**
Jun 2025 • Tracked and graded sponsor logos in live cricket with YOLOv8 for automated ROI reports.

Honors and Awards

- 2024 **Silver Medal**, *UET Taxila*, second highest aggregate in the session across all four years
2024 **Final Year Design Project Funding**, *Pakistan Engineering Council*, for PsycheSync
2022 to 2023 **Laptop Award**, *Prime Minister's Youth Laptop Scheme*, Pakistan
2018 to 2020 **1st Position in Pre Engineering and Full Tuition Waiver with Stipend**, *HSSC*

Technical Community & Leadership

- 2020 **Active Member**, *Google Developer Student Club (GDSC)*, UET Taxila
2024 • Organised workshops and led hands on sessions on TensorFlow, Firebase, and Flutter.
2020 **Active Member**, *Microsoft Learn Student Ambassadors (MLSA)*
2024 • Co organised talks and study jams on Azure and ML, and mentored peers on ML deployment.
2020 **Event Host**, *IEEE, UET Taxila Chapter*
2024 • Hosted seminars and panels connecting students with AI and software professionals.
2020 **General Secretary**, *Umeed-e-Subh*, UET Taxila
2024 • Student run welfare community that established a school for underprivileged children across from UET Taxila.
• Helped run daily classes and an evening IT drive teaching graphic design, basic coding, and video editing so children could earn while still in school.

Relevant Coursework

- Mathematics** Calculus, Linear Algebra and Differential Equations, Probability and Statistics
ML & Vision Machine Learning, Neural Networks, Computer Vision, Digital Image Processing
Systems Data Structures and Algorithms, Operating Systems, Databases, Cloud Computing

Technical Skills

- Programming** Python, C++, CUDA, Go, SQL, Unix Shell
ML PyTorch, TensorFlow, Diffusers, Transformers, PEFT, TorchAO, LoRA and QLoRA, Diffusion and Flow Models, LLMs, VLMs, Knowledge Distillation
Train & Infer FSDP, DTensor, Tensor and Context Parallelism, Quantization, Mixed Precision, FlashAttention, SageAttention, Sparse Attention, KV and Diffusion Caching, PyTorch Compilation
Systems NVIDIA Hopper, Blackwell, Ada, Ampere; AMD ROCm; Triton, CUTLASS, cuDNN, NCCL, vLLM, Docker, Kubernetes, AWS, RunPod
Vision YOLO, SAM, Detectron2, Tracking, Reidentification, Pose Estimation
Evaluation Nsight Systems, Latency and Throughput Benchmarking, FID, CLIP Score, SSIM, LPIPS

References

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